

FairSearch-arXiv: Evaluating and Mitigating Institutional Bias in Academic RAG

Project Update 1 (Week 6)

Ansh Kanungo
Northeastern University

Andrew Hannah
Northeastern University

Wan Chi Kao
Northeastern University

ABSTRACT

Retrieval-Augmented Generation (RAG) systems are becoming a primary gateway to scientific literature, yet the fairness properties of their retrieval components remain underexplored. We present an audit-and-mitigation study of *institutional homophily* in a dense-retrieval RAG pipeline over the arXiv cs.* corpus. This update reports our working Naive RAG baseline: from a corpus of 3.07M arXiv records we build a balanced 50K-abstract cs.* sample (2018–2025), index it with all-MiniLM-L6-v2 embeddings in ChromaDB, and evaluate retrieval over 18 curated neutral queries, achieving a mean Precision@10 of 0.71. We further report a key data-exploration finding: author-affiliation metadata is effectively absent from arXiv (0.3% coverage), which reshapes our Phase II plan for institution labeling toward external enrichment. We close with our roadmap for bias measurement and the fairness-aware re-ranking algorithm.

1 PROBLEM DESCRIPTION

Academic search powered by RAG embeds a user query, retrieves the top- K most similar abstracts via cosine similarity, and asks a large language model (LLM) to synthesize an answer. This pipeline concentrates two well-documented biases into a single opaque step: representational bias in pretrained embeddings, and consensus bias in generative summarization.

We study a specific manifestation of this concern, *institutional homophily*: the tendency of dense retrievers to over-rank papers from a small set of well-resourced (typically elite, Western) institutions, even when comparable work exists from regional or emerging research hubs. The mechanism is compounding. Embedding models are trained on corpora that over-represent elite-institution writing; papers from heavily-cited institutions cluster more tightly in embedding space, biasing nearest-neighbor retrieval; and the LLM then faithfully amplifies whatever distribution it receives, presenting an institutional imbalance to the user as the objective state of the field. The downstream effect is an algorithmic Matthew effect in science: the already-visible become more visible, while equally relevant work from underrepresented institutions is rendered invisible at the moment of information access.

2 RESEARCH QUESTIONS AND HYPOTHESES

RQ1 – Retrieval Parity. Does dense retrieval over arXiv cs.* systematically rank elite-institution papers above comparable regional-institution papers? **H1:** baseline retrieval will show a Selection Rate Ratio ≥ 2.0 favoring top-20 QS-ranked institutions across neutral queries.

RQ2 – Synthesis Neutrality. Does the LLM amplify institutional concentration during synthesis? We measure (a) *citation share* – the institutional distribution of *cited* papers versus *retrieved* papers

– and (b) *coverage rate* – the fraction of retrieved papers actually cited, split by institution group. **H2:** generated answers will over-represent elite-institution sources relative to the retrieved context. **RQ3 – Fairness–Utility Tradeoff.** When mitigation is applied, what is the tradeoff between fairness (Selection Rate Ratio) and utility (NDCG@10, Precision@10)? **H3:** a dual-criterion re-ranker will reduce institutional concentration with limited precision loss, tracing a smooth Pareto frontier as the fairness weight is swept.

3 BACKGROUND

Fair ranking has a mature literature; its intersection with RAG is recent. We review ten peer-reviewed works, organized as foundational methods (§3.1) and recent findings on bias in retrieval and RAG (§3.2).

3.1 Foundational Methods

Carbonell and Goldstein [1998]. Introduced Maximal Marginal Relevance (MMR) to reduce redundancy in retrieval and summarization, selecting each next document to maximize a λ -weighted blend of query relevance and novelty relative to already-selected documents. MMR is the canonical diversity baseline but optimizes *content* diversity, not demographic diversity – it is unaware of protected attributes such as author institution, the gap our work targets.

Singh and Joachims [2018]. Reframed fair ranking as exposure allocation: because attention follows a steep position-bias curve, rankings allocate a scarce resource. They defined Demographic Parity, Disparate Treatment, and Disparate Impact criteria and solved fair ranking as a linear program over doubly-stochastic matrices. The method requires predefined protected groups and a known position-bias model, and was evaluated on learning-to-rank data rather than academic search.

Karako and Mangala [2018]. Proposed FMRR, a fairness-aware MMR variant that redefines the similarity term so documents sharing a demographic group are treated as similar, spreading results across groups. It is the closest prior algorithm to our proposed extension, but was evaluated on stock photos and does not separate content diversity from group diversity as independently tunable penalties – which our work will.

Rekabsaz, Kopeinik, and Schedl [2021]. Studied bias in BERT-based neural rankers, showing they amplify rather than merely inherit bias. They introduced the FaiRR measurement framework (separating collection bias from ranker bias) and AdvBERT adversarial debiasing. Focused on gender bias with curated queries; extension to institutional attributes requires a new parallel query set, part of our contribution.

Kim and Diaz [2025]. Conducted the first systematic evaluation of fair ranking inside RAG pipelines (ICTIR ’25), addressing both ranking fairness and attribution fairness across many RAG configurations, and finding that fairness-aware retrieval often preserves or improves generation quality. Their fairness groups come from existing benchmarks rather than academic-search settings; we specialize their framework to scholarly RAG with institutional groups.

3.2 Recent Findings (2024–2025)

Dai et al. [2024], KDD ’24. “Neural Retrievers are Biased Towards LLM-Generated Content.” Show that dense retrievers systematically over-rank LLM-generated text over human-written text of equal relevance (“source bias”), and propose a debiasing constraint. Establishes that dense retrievers encode and amplify provenance signals — directly motivating our concern that they may likewise encode institutional signals.

Hu et al. [2024]. “No Free Lunch: Retrieval-Augmented Generation Undermines Fairness in LLMs, Even for Vigilant Users.” Demonstrate that adding retrieval can undermine an LLM’s fairness alignment even when the external corpus is censored/unbiased, via a three-level user-awareness threat model. Shows the synthesis stage itself can introduce bias — the premise of our RQ2.

“Measuring the Fairness Gap Between Retrieval and Generation in RAG” [SIGIR ’25]. Proposes a cognitive-complexity evaluation framework extending IR fairness metrics with centrality-based measures, quantifying how fairness disparities propagate from the retrieval phase into the generation phase across lexical (BM25) and dense (BGE) retrievers. Directly supports our RQ2 method of comparing retrieved vs. cited distributions.

“Correctness is not Faithfulness in RAG Attributions” [ICTIR ’25]. Argues that citation *correctness* (is the cited source real?) differs from *faithfulness* (does the text reflect the source?), and proposes evaluation that separates them. Informs how we interpret the inline-citation signal underlying our RQ2 citation-share metric.

“FairDiverse” [SIGIR ’25]. A comprehensive open toolkit for fairness- and diversity-aware IR, unifying multiple fairness objectives and re-ranking strategies under one framework. Provides reference implementations and metrics we can adopt for our Phase III mitigation and evaluation.

4 DATASET

We use the Cornell University arXiv Metadata snapshot (Kaggle), distributed as newline-delimited JSON with one record per submission. Streaming the full file, we measured **3,073,376 total records** (correcting earlier ≈ 2.5 M estimates). Each record exposes 14 fields including id, title, abstract, authors, authors_parsed, categories, doi, journal-ref, and update_date.

Filtering and sampling. We keep records whose primary category begins with cs. and whose year is 2018–2025, yielding **557,514** qualifying papers. From these we draw a **50,000**-abstract sample stratified across eight target subcategories so no single topic dominates downstream fairness measurement. We drop abstracts shorter than 50 characters and deduplicate on arXiv id; the final ChromaDB index holds **49,909** documents. Table 1 shows the balanced subcategory distribution.

Table 1: Subcategory distribution of the 50K cs.* sample (top buckets).

Subcat	Count	Subcat	Count
cs.CV	5,739	cs.IR	5,575
cs.LG	5,739	cs.SE	5,575
cs.CL	5,652	cs.DB	4,723
cs.AI	5,598	cs.CR	5,596

Table 2: Baseline Naive RAG retrieval quality (18 neutral queries).

Metric	Value
Mean Precision@10	0.7056
Mean Recall@10	0.0085
Queries evaluated	18
Indexed documents	49,909
Embedding model	all-MiniLM-L6-v2 (384-d)

Preprocessing. We normalize whitespace in titles and abstracts, extract the year from update_date, and take the primary category as the first token of categories.

Key finding — affiliations are effectively absent. A central requirement of our study is labeling each paper’s institution. Measuring institution-signal coverage over 50K recent cs.* papers, we found the authors_parsed affiliation slot populated only **0.3%** of the time, report-no 1.1%, doi 24.9%, and journal-ref 17.9%. Institution labels therefore *cannot* be drawn from arXiv metadata. We also tested the OpenAlex API (by DOI and by identifier) on sample papers and obtained empty institution lists / zero results. We conclude that institution enrichment is a substantive Phase II task (robust OpenAlex queries, a Semantic Scholar fallback, CrossRef for the 25% with DOIs, and author-name matching) rather than a simple lookup. This finding reshapes our methodology and is itself a contribution of this update.

5 EXPERIMENTATION AND RESULTS

System. Our Naive RAG baseline embeds abstracts with

sentence-transformers/all-MiniLM-L6-v2 (384-dim, cosine, normalized), indexes them in ChromaDB (local, persistent), and synthesizes answers with Gemini 1.5 Flash. A query is embedded, the top-10 abstracts are retrieved by cosine similarity, and the LLM is prompted to answer using only those abstracts with inline citations.

Evaluation protocol. Lacking human relevance judgments, we use a transparent, reproducible proxy. We issue 18 curated *neutral natural-phrases queries* (e.g., “Recent advances in graph neural networks”), each mapped to an expected cs.* subcategory. A retrieved paper is *relevant* if it (i) is in the expected subcategory and (ii) shares at least two distinctive terms with the query. Precision@10 is the relevant fraction of the top 10; Recall@10 is relevant-retrieved over total proxy-relevant in the corpus.

Results. Table 2 summarizes the baseline.

A mean **Precision@10 of 0.71** indicates the retriever surfaces relevant papers effectively for an untuned baseline — roughly seven of every ten retrieved papers are on-topic. Per-query precision

ranges from perfect (1.0 for graph neural networks, LLM text generation, and neural information retrieval) to strong (0.8–0.9 for object detection, semantic segmentation, question answering, software testing) to low (0.2 for “reinforcement learning for robotics control” and “distributed training of deep learning models”).

The low-scoring queries are instructive: they reflect arXiv *cross-filing* rather than retrieval failure. Most reinforcement-learning and distributed-training papers are filed under cs.LG rather than the cs.RO/cs.DC subcategories we assigned, so the strict subcategory rule marks otherwise-relevant results as non-relevant. This highlights the noisiness of author-assigned categories, a factor we control for by stratification.

On Recall@10. Recall is low by construction: each broad query has hundreds of proxy-relevant papers in the corpus while $K = 10$, capping achievable recall near 0.02 regardless of retrieval quality. We report it for completeness per the evaluation requirements but treat **Precision@10 as the primary quality signal**; a more informative recall would require a small human-labeled or known-item ground-truth set, which we note as future work.

6 PLAN FOR NEXT PHASE

Our immediate next phase targets the institution-labeling bottleneck and the bias audit it unblocks. (1) *Enrichment*: resolve arXiv IDs to author affiliations via robust OpenAlex queries plus a Semantic Scholar fallback and CrossRef for DOI-bearing papers, reporting coverage *by group* to detect non-random missingness (regional institutions may be labeled less often). (2) *Demographic mapping*: label institutions Elite vs. Regional using top-20 QS rankings, with CS h5-index and Global North/South robustness proxies. (3) *RQ1 audit*: compute Selection Rate Ratio over the neutral query set. (4) *RQ2*: measure citation-share and coverage-rate from Gemini inline citations. (5) *RQ3 mitigation*: implement a fairness-aware re-ranker extending MMR with an institutional-overlap penalty, sweep its weight, and plot the fairness–utility Pareto frontier. (6) *Dashboard*: a Streamlit interface comparing baseline and fair retrieval with a fairness scorecard.

7 CODE AVAILABILITY

All code, the evaluation query set, and baseline results are available at github.com/Anshkanungo/FairSearch, including the ingestion, indexing, RAG, and evaluation pipeline with a README.

REFERENCES

- [1] Jaime Carbonell and Jade Goldstein. 1998. The Use of MMR, Diversity-Based Reranking for Reordering Documents and Producing Summaries. In *SIGIR '98*. ACM, 335–336.
- [2] Ashudeep Singh and Thorsten Joachims. 2018. Fairness of Exposure in Rankings. In *KDD '18*. ACM, 2219–2228.
- [3] Chen Karako and Putra Manggala. 2018. Using Image Fairness Representations in Diversity-Based Re-ranking for Recommendations. In *UMAP '18 (Adjunct)*. ACM.
- [4] Navid Rekabsaz, Simone Kopeinik, and Markus Schedl. 2021. Societal Biases in Retrieved Contents: Measurement Framework and Adversarial Mitigation of BERT Rankers. In *SIGIR '21*. ACM, 306–316.
- [5] To Eun Kim and Fernando Diaz. 2025. Towards Fair RAG: On the Impact of Fair Ranking in Retrieval-Augmented Generation. In *ICTIR '25*. ACM, 33–43.
- [6] Sunhao Dai, Yuqi Zhou, Liang Pang, Weihao Liu, Xiaolin Hu, Yong Liu, Xiao Zhang, Gang Wang, and Jun Xu. 2024. Neural Retrievers are Biased Towards LLM-Generated Content. In *KDD '24*. ACM.
- [7] Mengxuan Hu, Hongyi Wu, Zihan Guan, Ronghang Zhu, Dongliang Guo, Daiqing Qi, and Sheng Li. 2024. No Free Lunch: Retrieval-Augmented Generation Undermines Fairness in LLMs, Even for Vigilant Users. arXiv:2410.07589.
- [8] 2025. Measuring the Fairness Gap Between Retrieval and Generation in RAG Systems Using a Cognitive Complexity Framework. In *SIGIR '25*. ACM.
- [9] 2025. Correctness is not Faithfulness in Retrieval-Augmented Generation Attributions. In *ICTIR '25*. ACM.
- [10] Chen Xu, Zhirui Deng, Clara Rus, Xiaopeng Ye, Yuanna Liu, Jun Xu, Zhicheng Dou, Ji-Rong Wen, and Maarten de Rijke. 2025. FairDiverse: A Comprehensive Toolkit for Fairness- and Diversity-Aware Information Retrieval. In *SIGIR '25*. ACM.